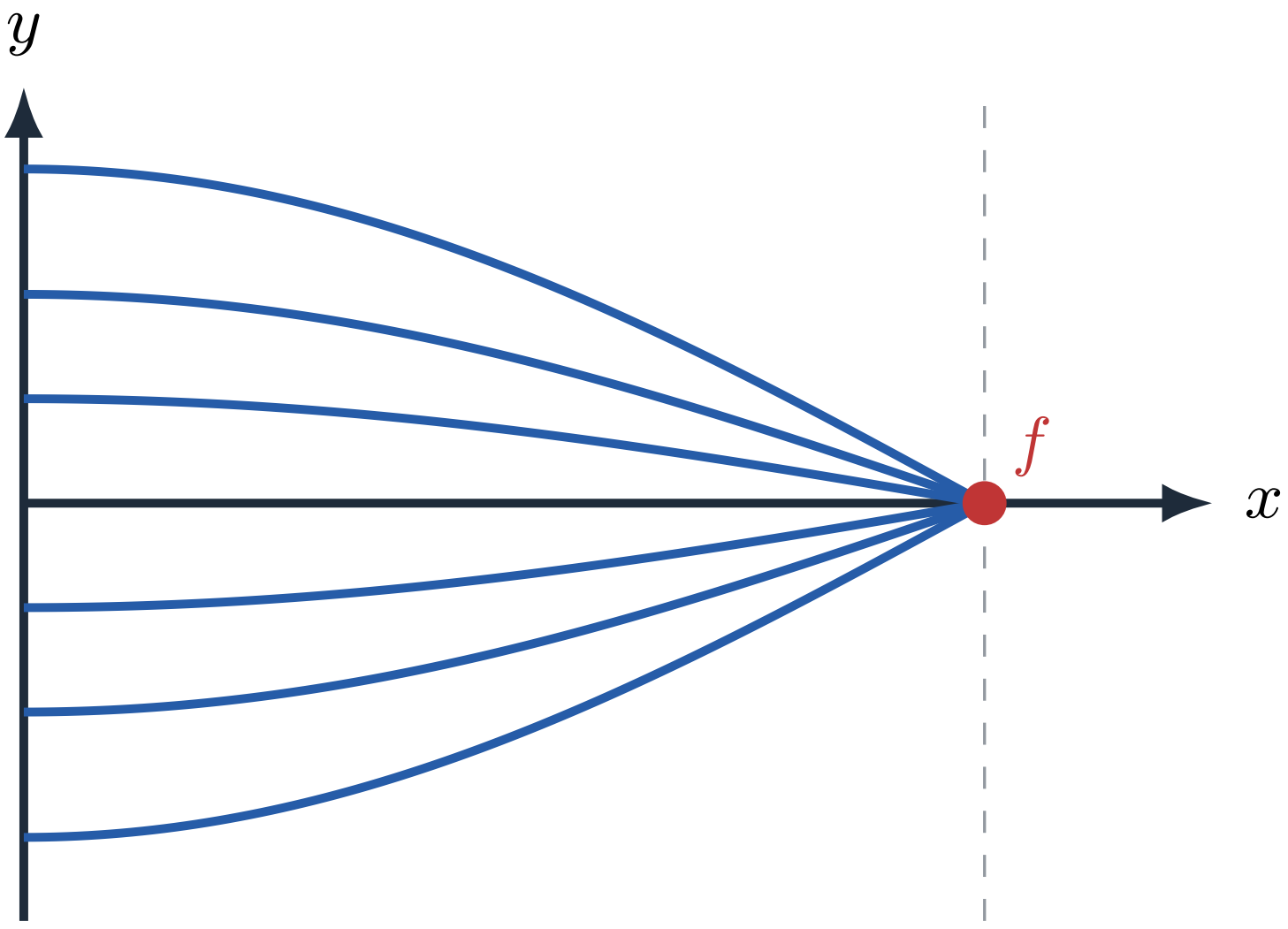

$$y'' + \frac{2(n^2-1)}{n\rho_0^2} y = 0$$

$$f/\rho_0 = \frac{\pi}{2}\sqrt{\frac{n}{2(n^2-1)}}$$